

shape that is actually curved. The term *convexity* describes price sensitivity to interest changes more precisely. Bond prices are more sensitive to rate decreases (prices rise more) than to equivalent rate increases (prices fall by less). These differences are accounted for by the fact that changing interest rates also impact duration. The duration for a given bond rises as interest rates fall, and future cash flows are discounted by less. But for a household retiree, duration provides a close enough approximation to this relationship, and only those with a greater interest in the mathematics of bond pricing should worry about further adding bond convexity to their discussion.

Though somewhat technical, this discussion of bond duration is important because the concept also applies to retirement spending liabilities and, therefore, the ability to meet retirement goals. Retirement spending has a duration that can be defined in the same way as an effective maturity for those cash flows. It is an average of when expenses must be paid, weighted by the size of the discounted values of those expenses.

Individual bonds have a duration. A bond fund, which is a collection of bonds, also has a duration equal to the average duration of each holding weighted by its proportion in the fund. Retirement liabilities have a duration, too. If the duration of the bonds and the spending liabilities can be matched to the same value, then the retiree has “immunized” his or her interest rate risk. Rising interest rates would lower the value of bond holdings, but rising rates also lower the present value of the future spending obligations. When durations are the same, both the asset and liability values are reduced by the same amount, and the retiree remains equally well-off in terms of the ability to meet liabilities. This is the meaning of immunizing interest rate risk.

However, if the durations do not match, then the retiree is exposed to interest rate risk. This is especially relevant for the case of holding bonds in bond funds. If the duration of the bond fund is less than the duration of the retirement liability, then rising interest rates may actually put the retiree into a better position, as the cost of retirement falls by more than the value of assets. On the other hand, if interest rates fall, the present value of cost-of-retirement liabilities increases by more than the value of the assets intended to fund those expenses, leaving the retiree in a worse position. For many retirees, the duration of their retirement liability may naturally tend to be higher than the